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Юрія

Кондратюка”

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Дисципліна «Технології розробки корпоративних Web-додатків»
Лабораторна робота № 3

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POST

http://localhost:5000/api/users/register

Send

DocsParamsAuthorizationHeaders (10)BodyScriptsSettings

Cookies

noneform-datax-www-form-urlencodedorawbinaryGraphQLJSON

SchemaBeautify

```
1 {
2   "name": "Vladlen Tymoshenko",
3   "email": "vladlenty@gmail.com",
4   "password": "password123",
5   "role": "admin"
6 }
```

BodyCookiesHeaders (7)Test Results

201 Created • 109 ms • 425 B •

JSON

PreviewVisualize

```
1 {
2   "success": true,
3   "data": {
4     "id": "69aac7e4cde99de7b83a06b1",
5     "name": "Vladlen Tymoshenko",
6     "email": "vladlenty@gmail.com",
7     "role": "admin",
8     "wishlist": [],
9     "createdAt": "2026-03-06T12:26:12.595Z"
10  }
11 }
```

POST

http://localhost:5000/api/users/login

Send

DocsParamsAuthorizationHeaders (10)BodyScriptsSettings

Cookies

noneform-datax-www-form-urlencodedorawbinaryGraphQLJSON

SchemaBeautify

```
1 {
2   "email": "vladlenty@gmail.com",
3   "password": "password123"
4 }
```

BodyCookiesHeaders (7)Test Results

200 OK • 86 ms • 570 B •

JSON

PreviewVisualize

```
1 {
2   "success": true,
3   "data": {
4     "id": "69aac7e4cde99de7b83a06b1",
5     "name": "Vladlen Tymoshenko",
6     "email": "vladlenty@gmail.com",
7     "role": "admin",
8     "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6IjY5YWYjN2U0Y2RlOTlkZTdldi00NhMDZlMSIsInJvbmUiOiJhZG1pbSImlhdCI6MTc3Mjg3MDAxX2xhbmIjoxNzc3NDA0OD0EwFQ.nPPM5j5v2tjxj3ESz-tpbj9vSmR0fx4u3e0fCFVYZtc"
9   }
10 }
```

POST

http://localhost:5000/api/products

Send

Docs

Params

Authorization

Headers (10)

Body

Scripts

Settings

Cookies

Headers

9 hidden

	Key	Value	Description		Bulk Edit	Presets
☒	Authorization	Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZiZ...				
	Key	Value	Description			

POST

http://localhost:5000/api/products

Send

Docs

Params

Authorization

Headers (10)

Body

Scripts

Settings

Cookies

none

form-data

x-www-form-urlencoded

raw

binary

GraphQL

JSON

Schema

Beautify

```
1 {
2   "name": "Intel Core i9-13900K",
3   "description": "High performance desktop processor",
4   "price": 589.99,
5   "category": "CPU",
6   "stock": 50,
7   "specs": {
8     "Cores": "24",
9     "Threads": "32",
10    "BaseClock": "3.0GHz",
11    "BoostClock": "5.8GHz"
12  }
13 }
```

Body

Cookies

Headers (7)

Test Results

201 Created • 15 ms • 566 B • •

JSON

Preview

Visualize

1

2

3

4

5

6

7

8

9

10

11

12

13

14

```
{
  "success": true,
  "data": {
    "id": "69aac852cde99de7b83a06b4",
    "name": "Intel Core i9-13900K",
    "description": "High performance desktop processor",
    "price": 589.99,
    "category": "CPU",
    "stock": 50,
    "images": [],
    "specs": {
      "Cores": "24",
      "Threads": "32",
      "BaseClock": "3.0GHz",
      "BoostClock": "5.8GHz"
    }
  }
}
```

GET

http://localhost:5000/api/products

Send

Docs

Params

Authorization

Headers (8)

Body

Scripts

Settings

Cookies

none

form-data

x-www-form-urlencoded

raw

binary

GraphQL

JSON

Schema

Beautify

1 GET /api/products HTTP/1.1

Body

Cookies

Headers (7)

Test Results

200 OK • 11 ms • 864 B • •

JSON

Preview

Visualize

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

```
{
  "success": true,
  "data": [
    {
      "id": "69a85e4a0f8d361fed8dc649",
      "name": "Intel Core i9-13900K",
      "description": "High performance desktop processor",
      "price": 589.99,
      "category": "CPU",
      "stock": 50,
      "images": [],
      "specs": {
        "Cores": "24",
        "Threads": "32",
        "BaseClock": "3.0GHz",
        "BoostClock": "5.8GHz"
      }
    },
    {
      "rating": 0,
      "createdAt": "2026-03-04T16:31:06.867Z"
    }
  ]
}
```

Приклади коду (фрагменти у звіті):

Визначення моделі API (DTO / Proto message / GraphQL Type).

```
class ProductDTO {  
  constructor(product) {  
    this.id = product._id;  
    this.name = product.name;  
    this.description = product.description;  
    this.price = product.price;  
    this.category = product.category;  
    this.stock = product.stock;  
    this.images = product.images;  
    this.specs = product.specs;  
    this.rating = product.rating;  
    this.createdAt = product.createdAt;  
  }  
}
```

```
module.exports = { ProductDTO };
```

```
class UserDTO {  
  constructor(user) {  
    this.id = user._id;  
    this.name = user.name;  
    this.email = user.email;  
    this.role = user.role;  
    this.wishlist = user.wishlist;  
    this.createdAt = user.createdAt;  
  }  
}
```

```
class AuthDTO {  
  constructor(user, token) {  
    this.id = user._id;  
    this.name = user.name;  
    this.email = user.email;  
    this.role = user.role;  
    this.token = token;  
  }  
}
```

```
module.exports = { UserDTO, AuthDTO };
```

Реалізація методу (Controller Action / Resolver / gRPC Service method).

```
const productService = require('./services/ProductService');
```

```
const { toProductDTO, toProductDTOList } =
```

```
require('./mappers/productMapper');
```

```
class ProductController {
```

```
  async createProduct(req, res) {
```

```
    try {
```

```
      const product = await productService.createProduct(req.body);
```

```
      res.status(201).json({ success: true, data: toProductDTO(product) });
```

```
    } catch (error) {
```

```
      res.status(400).json({ success: false, message: error.message });
```

```
    }
```

```
  }
```

```
  async getProductById(req, res) {
```

```
    try {
```

```
      const product = await productService.getProductById(req.params.id);
```

```
      res.status(200).json({ success: true, data: toProductDTO(product) });
```

```
    } catch (error) {
```

```
      res.status(404).json({ success: false, message: error.message });
```

```
    }
```

```
  }
```

```
  async getAllProducts(req, res) {
```

```
    try {
```

```
      const products = await productService.getAllProducts(req.query);
```

```
      res.status(200).json({ success: true, data: toProductDTOList(products) });
```

```
    } catch (error) {
```

```
      res.status(500).json({ success: false, message: error.message });
```

```
    }
```

```
  }
```

```
  async getProductsByCategory(req, res) {
```

```
    try {
```

```
      const products = await
```

```
productService.getProductsByCategory(req.params.category);
```

```
      res.status(200).json({ success: true, data: toProductDTOList(products) });
```

```
    } catch (error) {
```

```
      res.status(404).json({ success: false, message: error.message });
```

```
    }
```

```
  }
```

```
  async searchProducts(req, res) {
```

```
    try {
```

```

        const products = await productService.searchProducts(req.query.q);
        res.status(200).json({ success: true, data: toProductDTOList(products) });
    } catch (error) {
        res.status(400).json({ success: false, message: error.message });
    }
}

async updateProduct(req, res) {
    try {
        const product = await productService.updateProduct(req.params.id,
req.body);
        res.status(200).json({ success: true, data: toProductDTO(product) });
    } catch (error) {
        res.status(400).json({ success: false, message: error.message });
    }
}

async deleteProduct(req, res) {
    try {
        await productService.deleteProduct(req.params.id);
        res.status(200).json({ success: true, message: 'Product deleted successfully'
});
    } catch (error) {
        res.status(404).json({ success: false, message: error.message });
    }
}
}

module.exports = new ProductController();

```